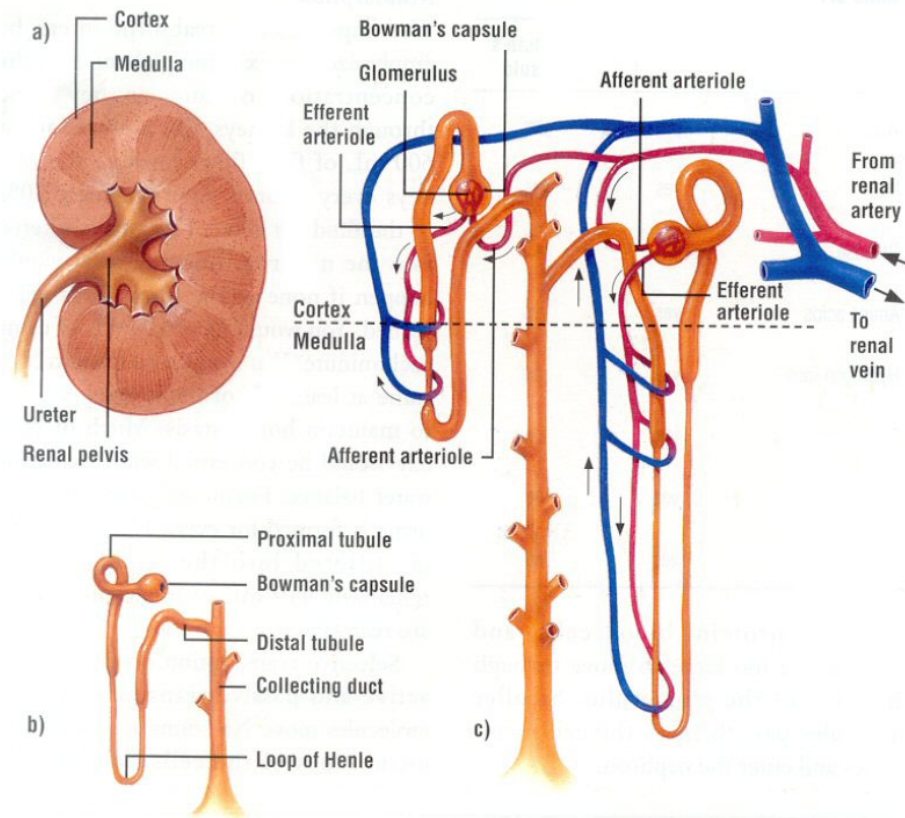


Glomerular and Tubulo-interstitial Diseases & Acute Kidney Injury

Sydney Tang

Renal parenchyma

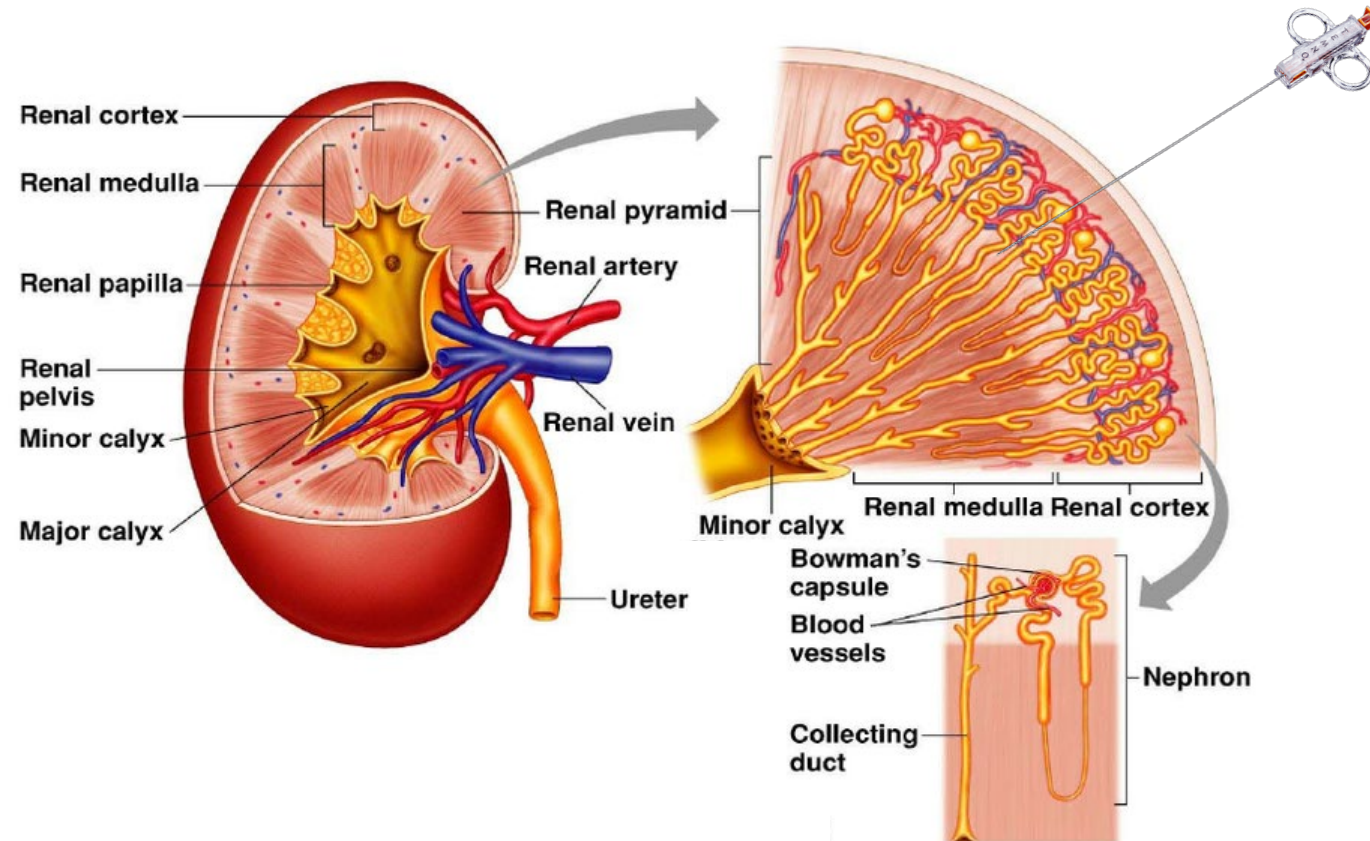
- the functional tissue of the kidney, consisting of the nephrons



KDIGO GN Guideline 2021

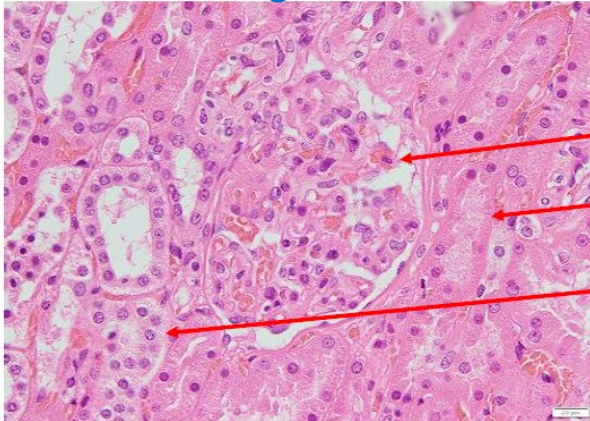
Practice Point 1.1.1: The kidney biopsy is the “gold standard” for the diagnostic evaluation of glomerular diseases.

However, under some circumstances, treatment may proceed without a kidney biopsy confirmation of diagnosis



Kidney histology

H&E staining



Cortex

Glomerulus

Proximal tubule

Distal tubule



Medulla

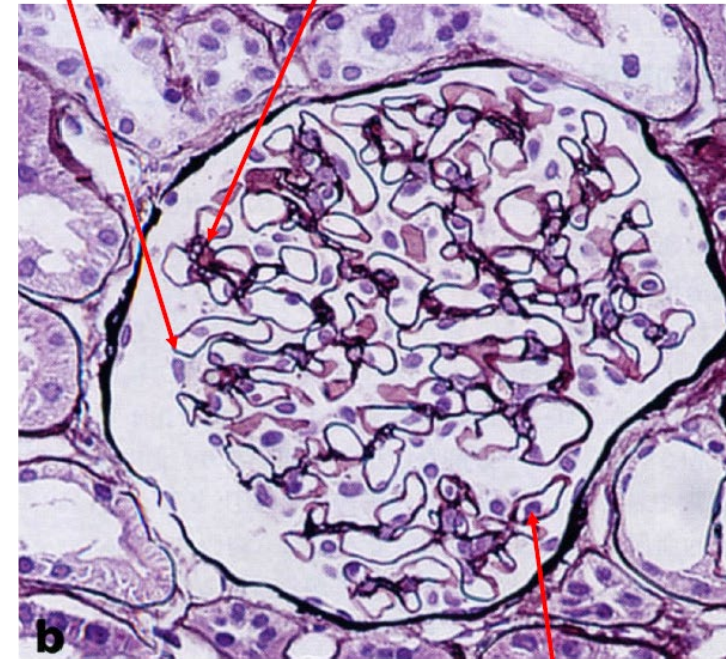
Collecting duct

Henle's loop

Silver staining

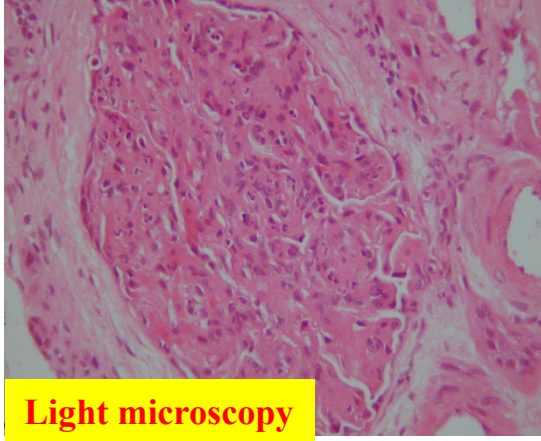
Podocyte

Mesangial cells

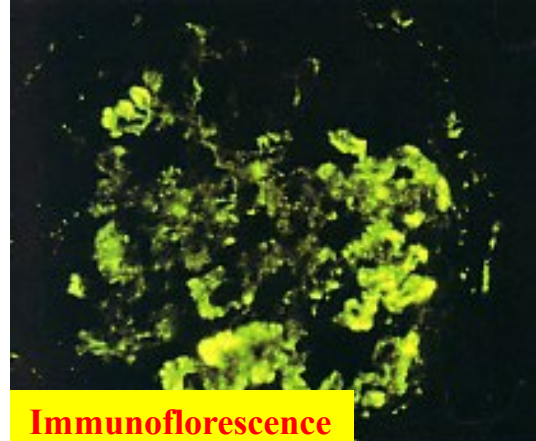


Endothelial cell

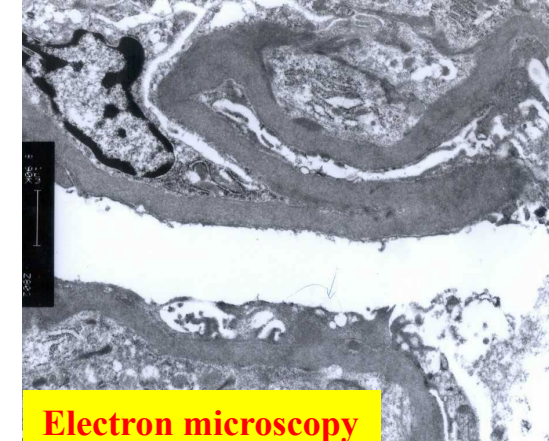
Standard histopathologic processing of kidney biopsies



Haematoxylin and eosin
Periodic acid-Schiff
Periodic acid silver-methenamine
Masson Trichrome
Congo red
Enzyme techniques



C': C1q, C3, C4, C4d
IgA, Gd-IgA1
IgG & subclasses
IgM
Kappa, Lambda

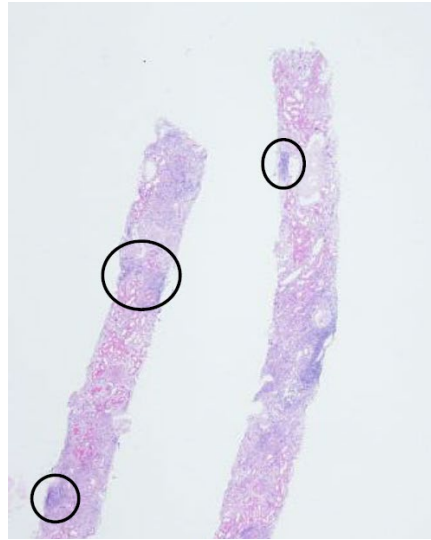


Ultrastructural evaluation:

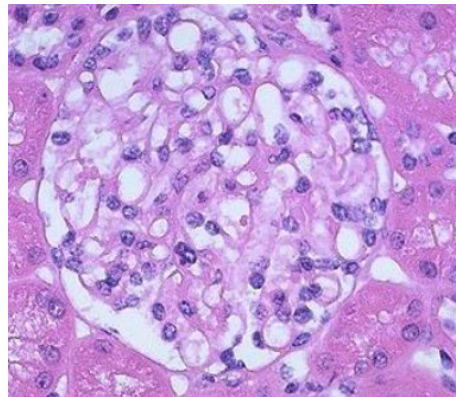
- Foot processes (podocytes)
- Basement membrane thickness
- Electron-dense deposits (EDDs)

H&E Staining

- Nuclei are stained **dark blue**
- Cytoplasm is stained **eosinophilic**
- Overall survey:
 - where is the lesion?
 - are cells abundant or scarce?
- Detailed assessment not possible



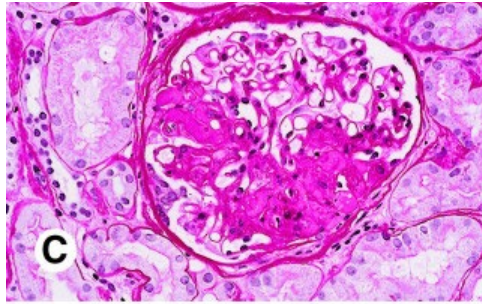
Normal



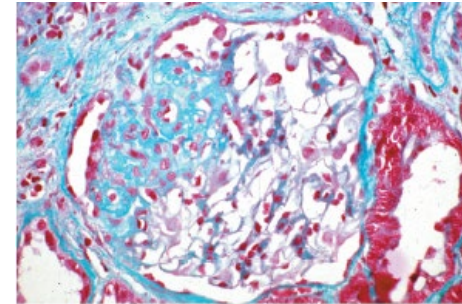
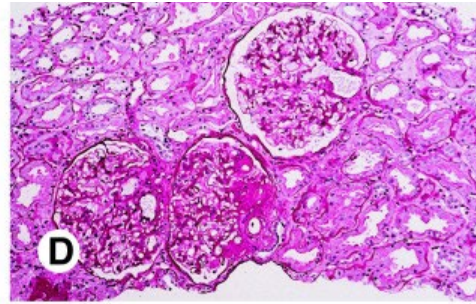
Hyalinosis



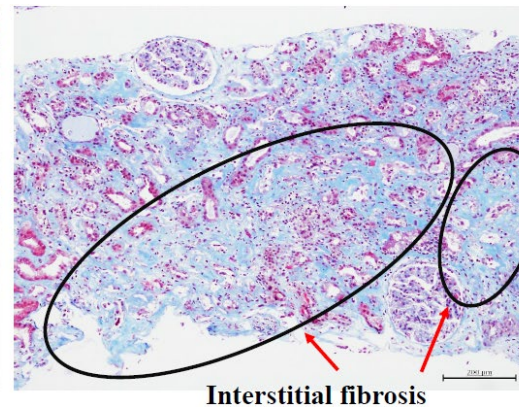
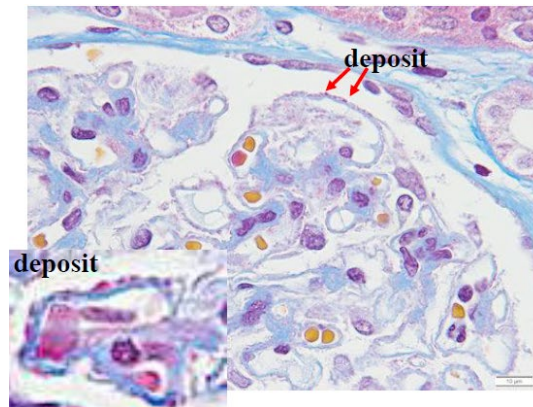
Focal Segmental Glomerulosclerosis (FSGS)



Periodic acid-Schiff: stains basement membrane and PAS-positive materials such as proteins and glycogen



Masson-Trichrome: red/blue/orange
Collagen are stained blue for evaluation of kidney fibrosis
Deposits are stained orange to red



Classification of glomerulonephritis by glomerular cellularity

Proliferative

Nonproliferative

Primary glomerular diseases

IgA nephropathy

Minimal change nephrotic syndrome

Membranoproliferative GN

Membranous nephropathy

Crescentic GN

Focal segmental glomerulosclerosis

IgM nephropathy

Thin membrane disease

Classification of glomerulonephritis by glomerular cellularity

Proliferative

Nonproliferative

Secondary glomerular diseases

Lupus nephritis
Poststreptococcal GN
Hepatitis B or C-related
membranoproliferative GN
Systemic vasculitis
Goodpasture syndrome

Diabetic nephropathy
Hypertensive nephrosclerosis
Amyloidosis
Light/heavy chain deposition disease
Alport's syndrome
Infection-related membranous
glomerulopathy
HIV nephropathy (FSGS or collapsing
glomerulopathy)
Drug-induced glomerulopathy
Malignancy-associated nephropathy
Reflux nephropathy

Classification of Glomerular Disease

- Usually by different patterns of histologic injury seen on biopsy
- No single classification is ideal since it cannot always be assumed that one histologic pattern has a single etiology (*e.g. the membranous pattern can be idiopathic, or due to lupus or viral hepatitis infection*) or that it will have a single clinical presentation (*e.g. IgAN may present with asymptomatic microscopic hematuria or synpharyngitic gross hematuria or RPGN*)
- Further, one etiology may produce a variety of histologic patterns (*e.g. lupus nephritis, IgAN, HBV infection*)

Presenting clinical features of renal parenchymal disorders

Clinical Presentation of Glomerular Disease:

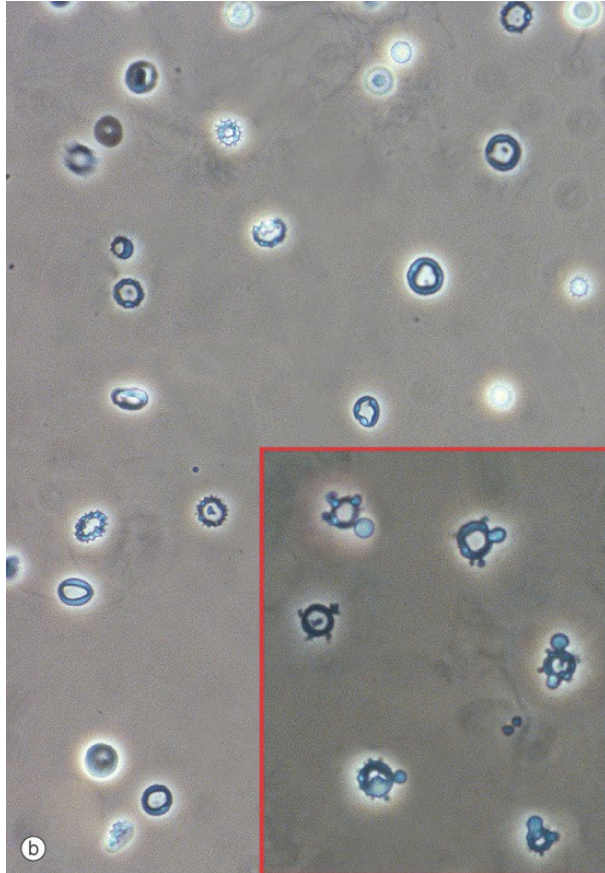
6 clinical syndromes

- Asymptomatic microscopic hematuria / proteinuria
- Macroscopic hematuria
- Acute nephritic syndrome
- Nephrotic syndrome
- Rapidly progressive glomerulonephritis
- Chronic glomerulonephritis / CKD

1. Asymptomatic Haematuria

- Either macroscopically or microscopically (>2 RBC/HPF in spun urine) detected blood in the urine, usually dysmorphic
- Detected through health check, pre-employment, pre-school, etc.
- Usually with normal glomerular filtration rates and no evidence of a systemic disease
- Variable degrees of proteinuria may be present, usually < 1 g/day

Dysmorphic RBCs of glomerular hematuria



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The dysmorphism consists mainly in irregularities of the cell membrane. *Inset*, Acanthocytes with their typical ring-formed cell bodies with one or more blebs of different sizes and shapes



RBC cast

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Macroscopic Hematuria

- Episodic painless macroscopic hematuria associated with glomerular disease is often brown or “smoky” rather than red, and blood clots are unusual

Differential diagnosis

- Urologic conditions
 - Stones
 - Tumour
- Renal conditions
 - Glomerulonephritis
 - Acute interstitial nephritis
 - Polycystic kidney disease
- Infection
 - Cystitis
 - TB
 - Schistosomiasis

Glomerular causes

- IgAN
- Alport's syndrome (hereditary nephritis)
 - Hearing deficit common
- Thin membrane disease (benign familial hematuria)

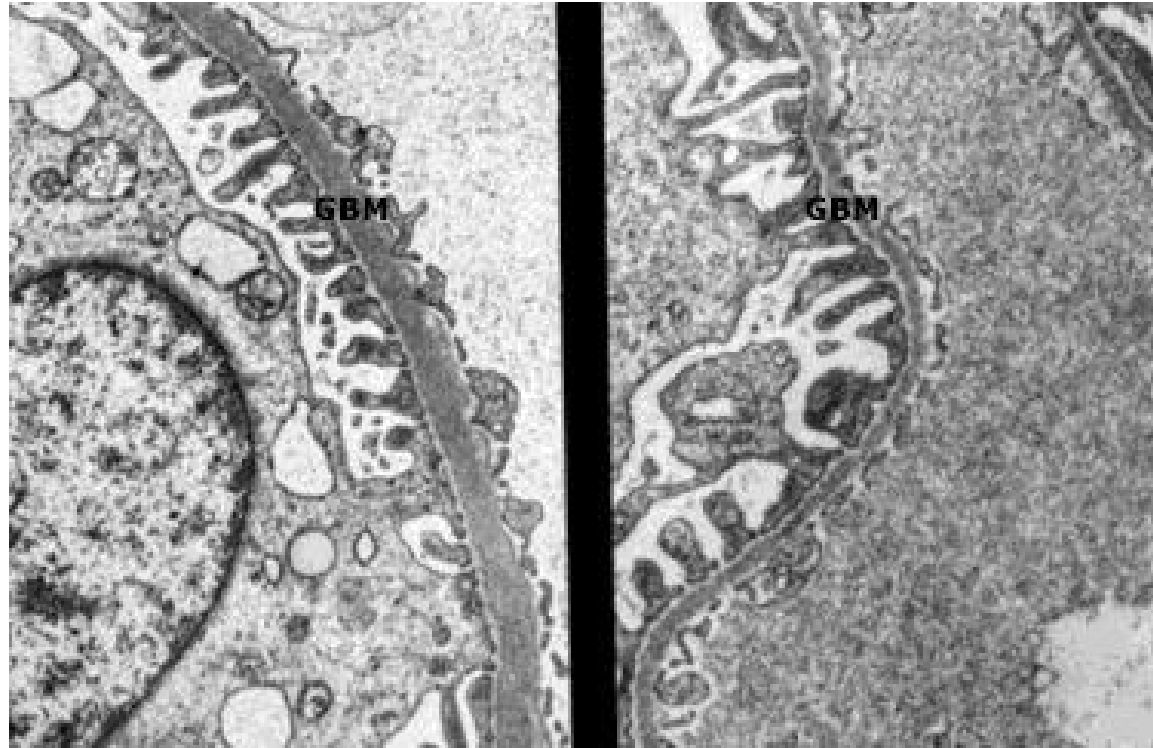
Alport syndrome

- Mutations type IV collagen genes
- Cochlea — ↓ adhesion of the Organ of Corti (the auditory sensory cells) to the basilar membrane via the defective alpha-3-4-5 type IV collagen
- X-linked in 80%
- Father-to-son transmission does not occur
- Women with X-linked Alport syndrome are heterozygous carriers of the disease mutation. Almost all have some degree of hematuria, and some develop renal failure [lyonization]

Evaluation

- The 3 conditions can often be distinguished by history / urinalysis of family members
- Gross hematuria
 - common in IgAN and Alport syndrome
 - unusual (< 10 %) in TBMD
- Family history of chronic kidney disease
 - Alport syndrome: renal failure / deafness (primarily in males); X-linked dominant mode of inheritance
 - IgAN: runs in families but no particular mode of inheritance

TBM disease

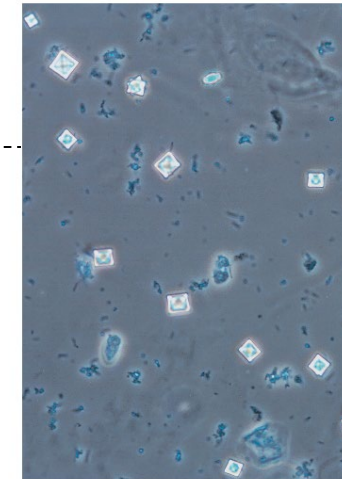
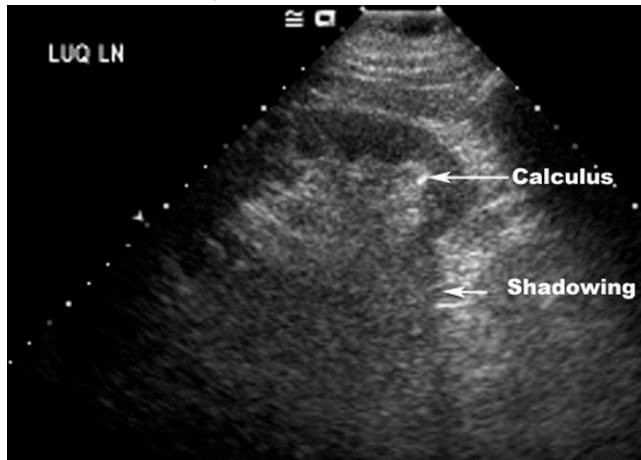


Normal: 300 to 400 nM

TBMD: 150-225 nM

Investigations for hematuria

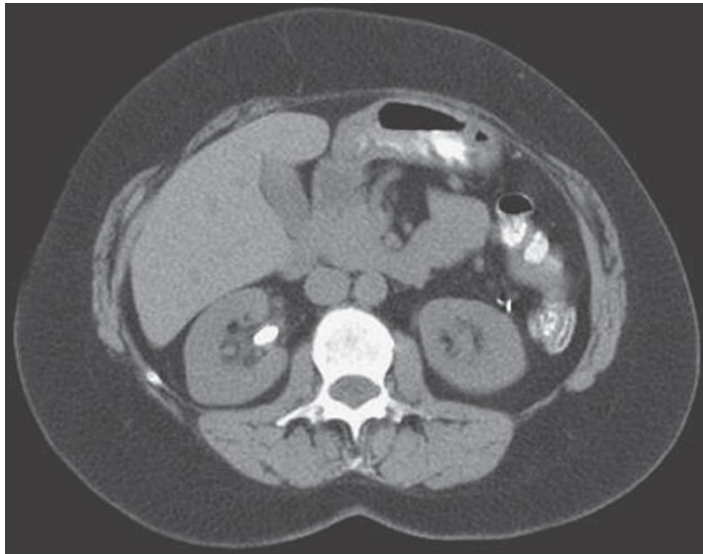
- Renal function
- Urine culture / cytology / AFB
- Urinalysis
- Urine microscopy
- KUB
- USG / Doppler



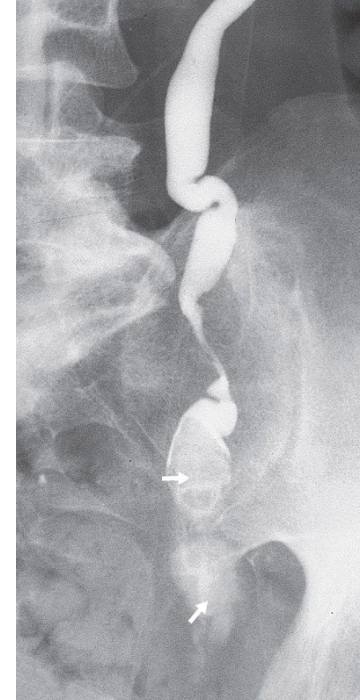
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Other Ix

- Cystoscopy
- IVP / RP
- CT urogram (radiation exposure 2 x IVP)

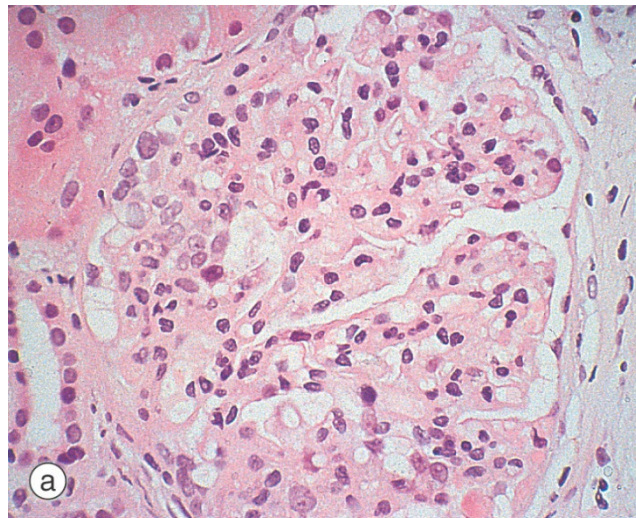


(From <http://www.gifhealthcare.com/userfiles/products/urologyimagegallery.html>.)



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- Pure tone audiometry
- Kidney biopsy
 - Usually not required in isolated haematuria
 - Indicated with concomitant significant proteinuria, e.g. $> 1 \text{ g / day}$ or $\text{UPCR} > 1 \text{ mg/mg}$ or 113 mg/mmol (such threshold should be lowered in selected patients)
 - Treatment may proceed without a biopsy in some conditions



Diffuse mesangial hypercellularity of IgAN (hematoxylin-eosin x300)

2. Isolated Proteinuria

- normal < 150 mg (usually 40 to 80 mg), of which approximately 10 mg is albumin.
- Urine albumin:
 - N < 20 mg/day (15 µg/min)
 - persistent microalbuminuria: 30 - 300 mg/day (20 to 200 µg/min)
 - Values above 300 mg/day (200 µg/min) are considered to represent overt proteinuria or macroalbuminuria, the level at which the standard dipstick becomes positive

Evaluation of proteinuria

- Dipstix
- Spot urine
- Timed urine (24 h)

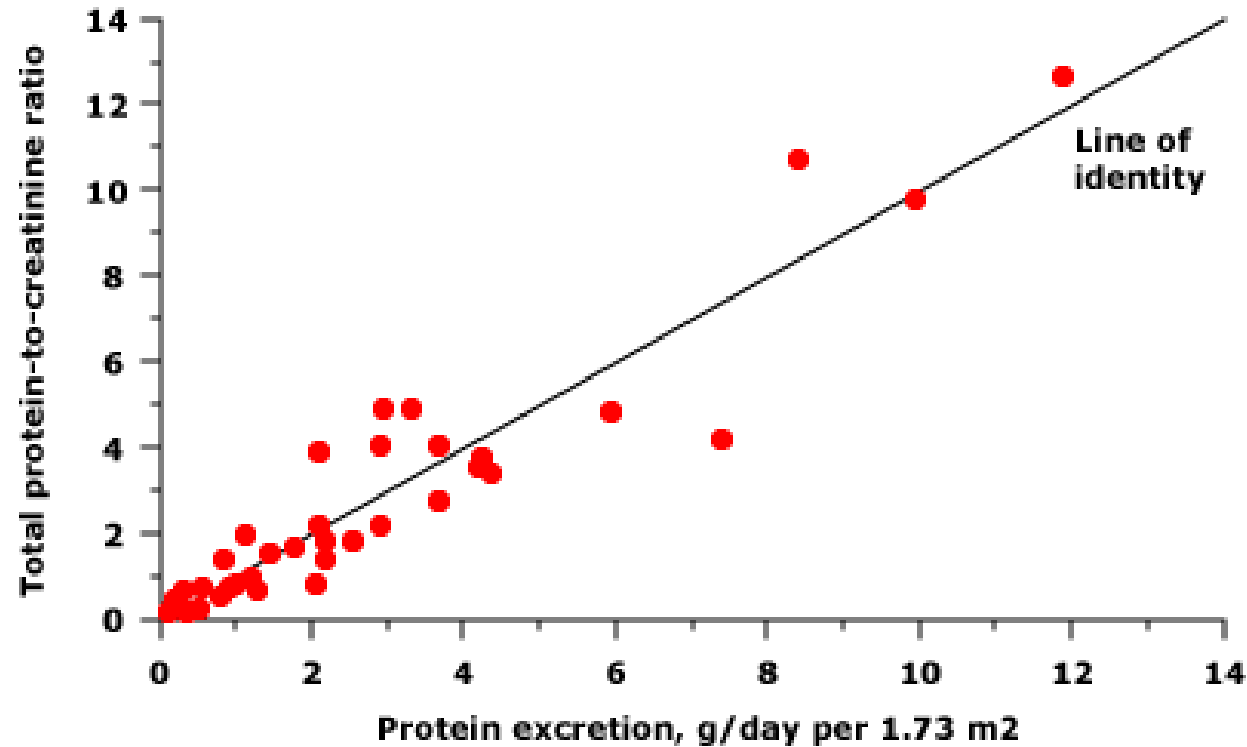
Standard urinary dipstick

- Measures albumin concentration via a colorimetric reaction between albumin and tetrabromophenol blue producing different shades of green according to the concentration of albumin in the sample.
 - Negative
 - Trace — between 15 and 30 mg/dL
 - 1+ — between 30 and 100 mg/dL
 - 2+ — between 100 and 300 mg/dL
 - 3+ — between 300 and 1000 mg/dL
 - 4+ — >1000 mg/dL



- The normal urine protein-to-creatinine ratio is less than 0.2 mg/mg
- Ratios greater than 3.5 mg/mg are in the nephrotic range for proteinuria

Protein-creatinine ratio to estimate protein excretion



Evaluation of proteinuria

- First morning specimens are preferred, but random specimens are acceptable
- *Screening* with urine dipsticks is acceptable for detecting proteinuria
- Patients with a positive dipstick test (1+ or greater) should undergo *confirmation* by a quantitative measurement
- Patients with two or more positive quantitative tests temporally spaced by 1 to 2 weeks should be diagnosed as having persistent proteinuria and undergo *further evaluation*
- *Monitoring* proteinuria in patients should be performed using quantitative measurements

CKD Prognosis by albuminuria and eGFR – from NKF and KDIGO

	A1 Normal to Mild Albuminuria <30 mg/g <3 mg/mmol	A2 Moderate Albuminuria 3 - 300 mg/g 3 - 30 mg/mmol	A3 Severely High Albuminuria >300 mg/g >30 mg/mmol
Stage 1 (eGFR ≥ 90)	Low Risk	Moderate Risk	High Risk
Stage 2 (eGFR 60 - 89)	Low Risk	Moderate Risk	High Risk
Stage 3a (eGFR 45 - 59)	Moderate Risk	High Risk	Very High Risk
Stage 3b (eGFR 30 - 44)	Moderate Risk	Very High Risk	Very High Risk
Stage 4 (eGFR 15 - 29)	Very High Risk	Very High Risk	Very High Risk
Stage 5 (eGFR < 15)	Very High Risk	Very High Risk	Very High Risk

Low Risk

Moderate Risk

High Risk

Very High Risk

Pitfalls in proteinuria assessment

False positive results

- Strenuous exercising within 24 hours
- Concomitant systemic infection
- UTI
- Fever
- Pregnancy
- Marked hypertension
- Haematuria

Orthostatic proteinuria

- Increased protein excretion in the upright position, but not when supine
- Benign condition in young subjects
- Mechanism unclear, ? neurohumoral activation
- Proteinuria generally < 1 g/day, but may exceed 3 g/day
- Requires no further evaluation or specific therapy. In many patients, the condition resolves over time
- Split urines are collected to identify this condition

3. Acute nephritic syndrome

- Abrupt onset of macroscopic hematuria
- Oliguria
- Acute renal failure
- Fluid retention, manifested by edema and hypertension
- Urinary protein < 3 g/d

Poststreptococcal glomerulonephritis

- Predominantly affects children between the ages of 2 and 10 years
- Pathogenesis: streptococcal antigens “planted” in the glomerulus during early phase of streptococcal infection, followed 10 to 14 days later by a host immune response

Clinical picture

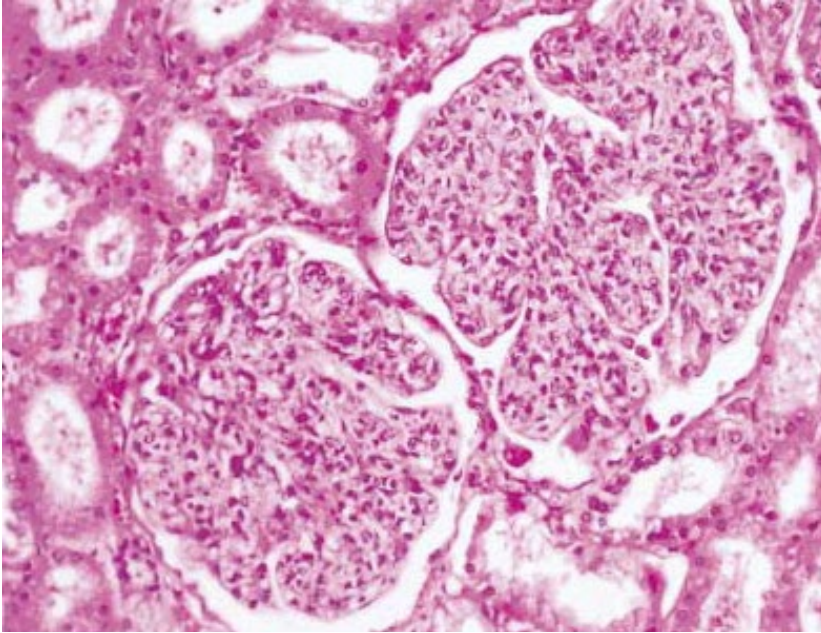
- Acute, reversible disease characterized by spontaneous recovery in the vast majority of patients.
- Typically, gross hematuria, oliguria, HT and edema develop 7 days to 12 weeks after the streptococcal infection (throat/skin).
- Spontaneous resolution of the clinical manifestations is generally rapid: diuresis usually ensues within 1-2 weeks, and renal function returns to baseline within 4 weeks.

Ix

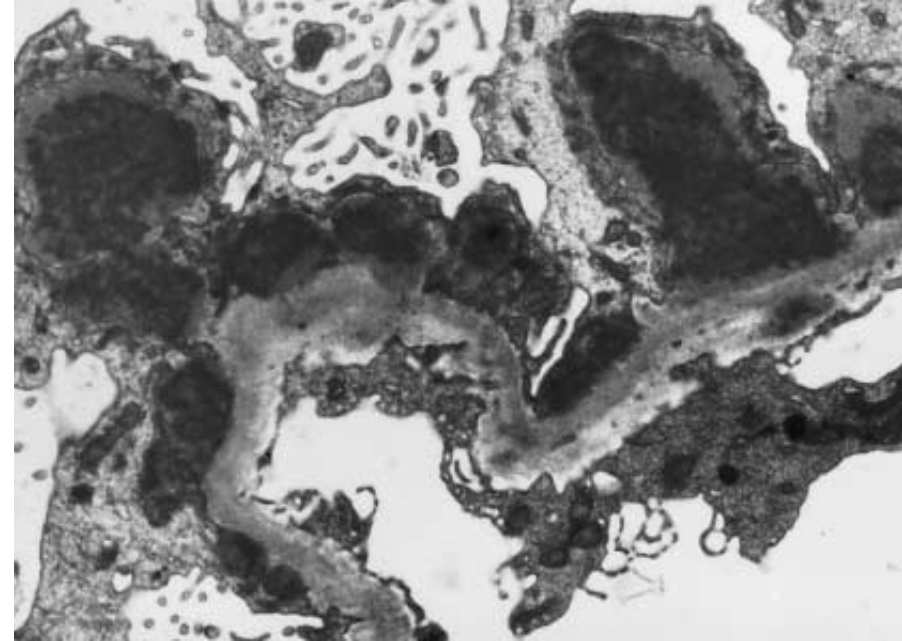
- Total hemolytic complement activity and C3 concentrations are depressed
- Antibodies to streptolysin O may be raised
 - Not all strains produce streptolysin O
- Note: Persistently low C3 > 8 weeks raises the possibility of lupus nephritis or membranoproliferative glomerulonephritis.

Treatment

- Depends on etiology
- For acute poststreptococcal Gn, penicillin may be indicated
- Otherwise, supportive therapy +/- temporary dialysis may be needed



diffuse hypercellularity x156



subepithelial deposits referred to
as "humps" x9350

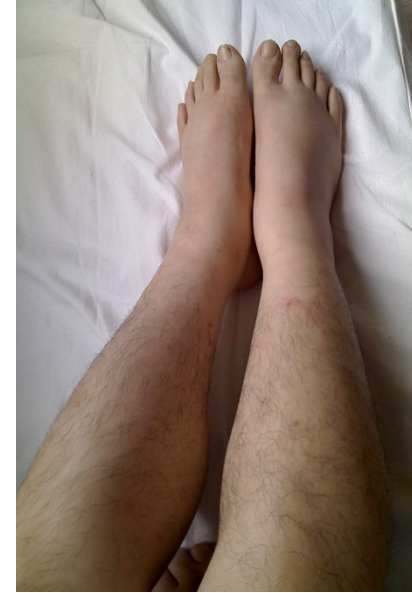
4. The Nephrotic Syndrome

- Massive proteinuria (>3.5 g per day or 40 mg/h/m² in children)
- Generalized edema
- Hypoalbuminemia (<30 g/L)
- Hyperlipidemia and lipiduria (varying degrees)

Clinical features



Periorbital edema in the early morning in a nephrotic child. The edema resolves during the day under the influence of gravity



Dependent ankle edema in a young man with **frothy urine**



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Muehrcke's bands in nephrotic syndrome. The white band grew during a transient period of hypoalbuminemia caused by the nephrotic syndrome



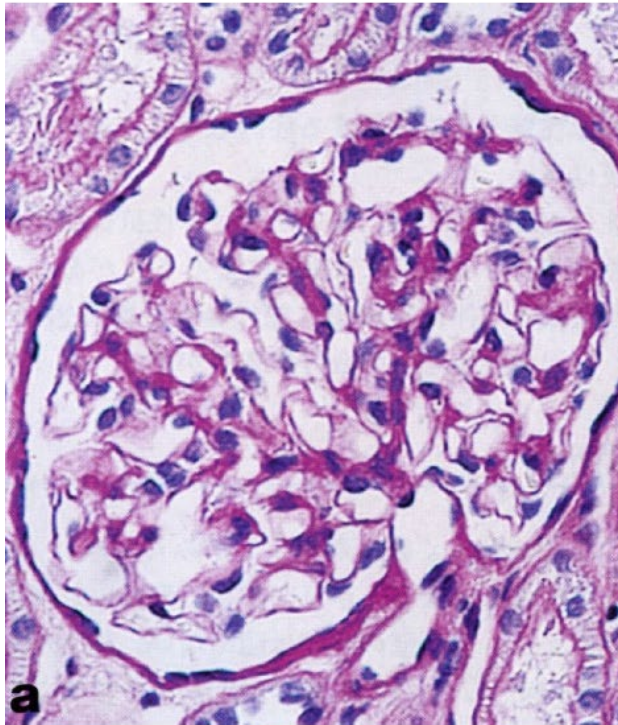
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Xanthelasma

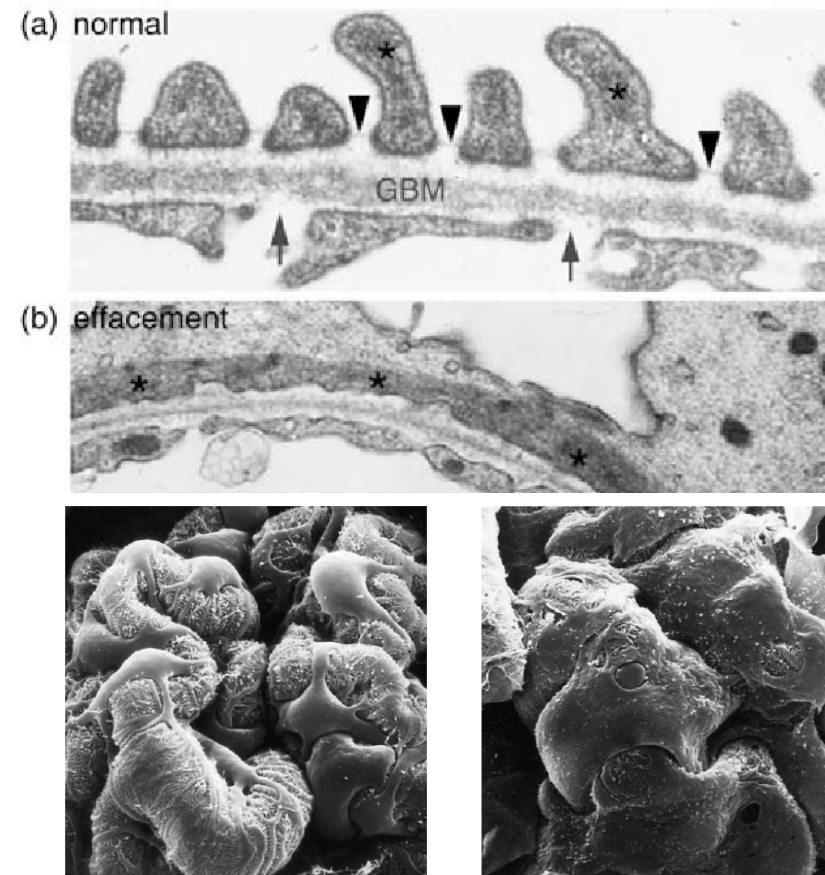
Primary kidney diseases

- Focal segmental glomerulosclerosis
- Membranous glomerulopathy
- Minimal change disease
- Membranoproliferative glomerulonephritis
- IgA nephropathy (occasionally)

Kidney biopsy



Podocyte foot process effacement



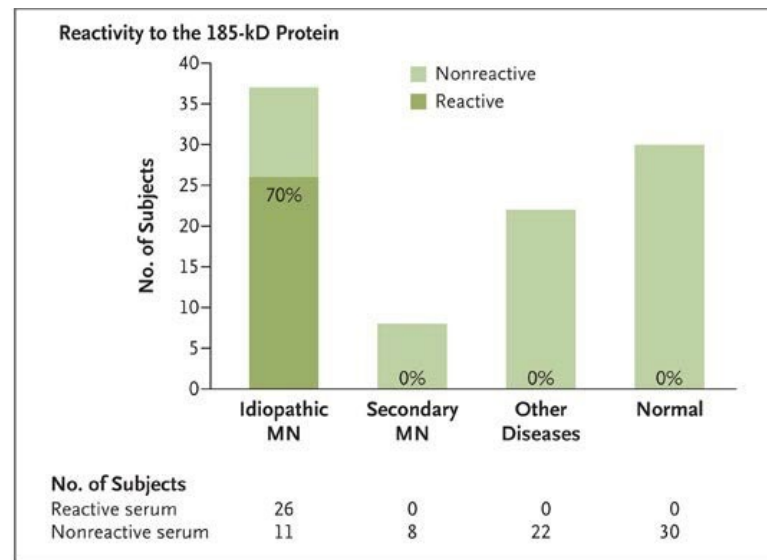
Systemic diseases

- Diabetes mellitus
- Systemic lupus erythematosus
- Secondary membranous nephropathy
 - Infection: HBV, HCV, HIV, malaria, syphilis
 - Malignancy: lymphoma, adenocarcinoma (lung, GI tract, breast), myeloma
 - Drugs: Penicillamine, gold, mercury, NSAIDs, probenecid
- Amyloidosis

Evaluation

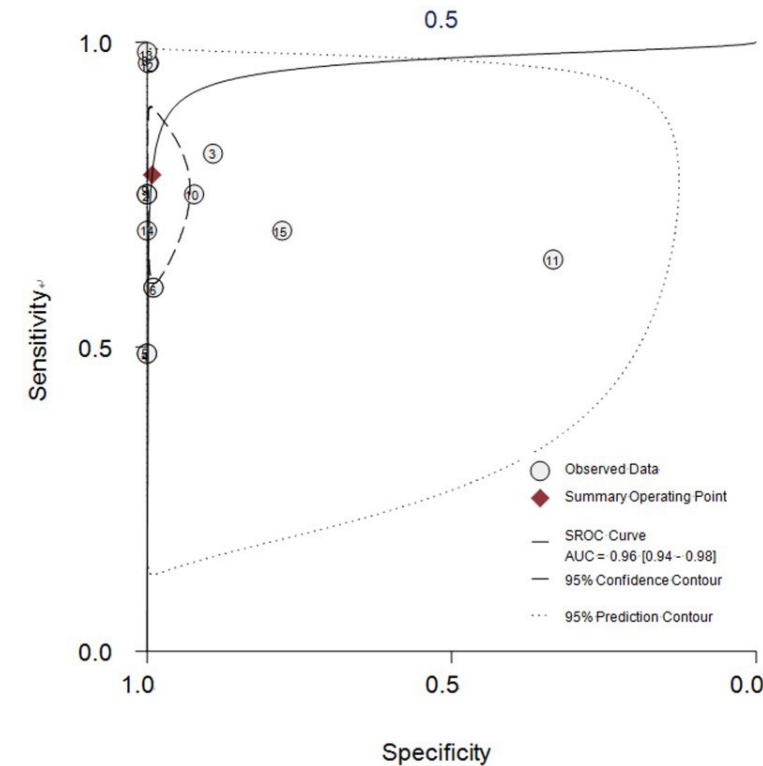
- Full blood counts
- Renal biochemistry
- Urine protein quantification
- FG
- Immune markers
- Ig pattern
- HBsAg
- Anti-HCV
- Anti-HIV
- Anti-PLA2R
- Tumour screening: CXR, Stool OB, other relevant markers

Discovery of the M-type phospholipase A2 receptor as a target antigen in idiopathic membranous nephropathy



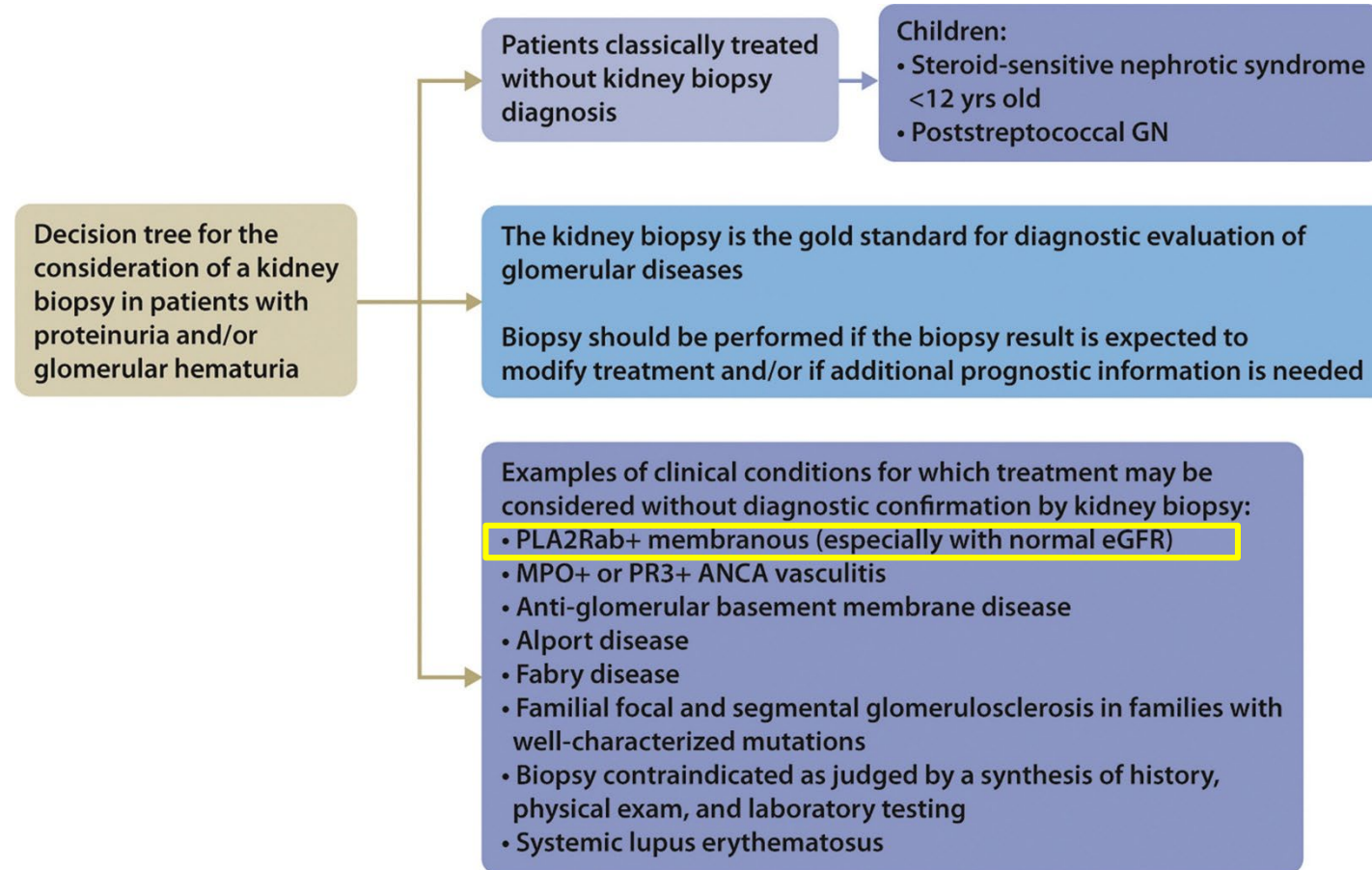
Beck LH, *et al.* NEJM 2009

Diagnosis value of iMN by anti-PLA2R



Du Y, *et al.* Plos One 2014

Conditions for which treatment can be considered without a kidney biopsy

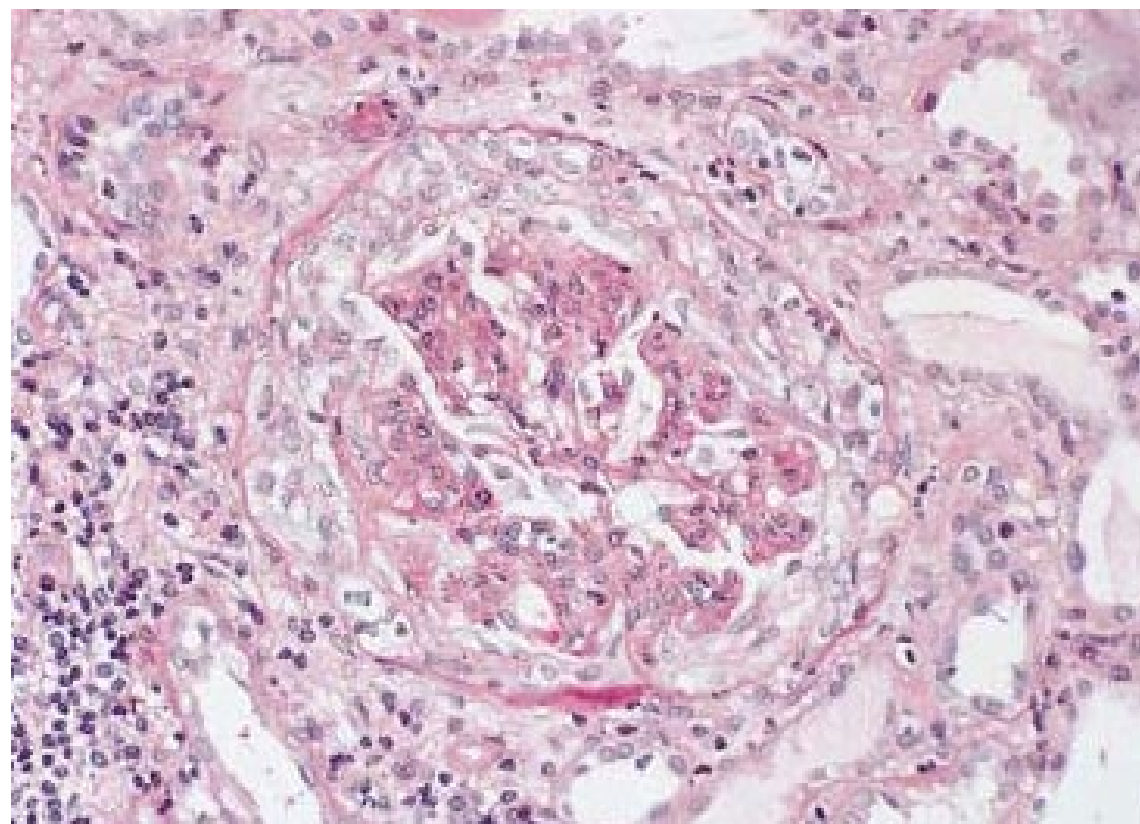


Treatment

- Corticosteroid remains the mainstay of treatment
- Second-line therapy such as calcineurin inhibitor may be useful for steroid-resistant or steroid-dependent cases to achieve steroid-sparing effects
- Other forms of treatment

5. Rapidly Progressive Glomerulonephritis (RPGN)

- Signs of glomerulonephritis (haematuria, proteinuria, and red-cell casts)
- Rapid decline in renal function that can lead to kidney failure within days to weeks
- Pathological hallmark of this syndrome is the presence of cellular crescents
- Beware of other organ system involvement, e.g. pulmonary haemorrhage (Goodpasture's) or hepatic failure (leptospirosis)



Etiology

- Pauci-immune Gn (systemic vasculitis)
- Anti-GBM disease / Goodpasture syndrome
- Crescentic IgAN
- Acute postinfectious Gn

Multisystem diseases

- Systemic vasculitides
- Cryoglobulinemia
- Systemic lupus erythematosus

Ix

- Full blood counts
- Biochemistry
- Urine protein quantification
- Urine sediments
- Autoimmune markers
 - ANCA, Anti-GBM, ANA, Anti-dsDNA, C3/C4
- CXR
- DLCO

Treatment

- Tailored according to etiology
- Usually requires systemic corticosteroid +/- immunosuppressants/biologics
- Plasmapheresis if anti-GMB Ab present

6. Chronic Glomerulonephritis

- A syndrome manifested by progressive kidney function loss in patients with glomerular inflammation, proteinuria, haematuria, and often, hypertension
- An important cause of chronic kidney disease (CKD)
- The kidneys show variable degrees of shrinkage (atrophy)

Evaluation

- Ascertain the chronic nature of the disease
 - Full blood counts: NcNc anaemia
 - Renal biochemistry: $\uparrow$ serum urea, creatinine, PO_4 , $\downarrow$ Ca +/- $\uparrow$ K+ and metabolic acidosis
 - Urine protein quantification
 - USG
 - Kidney biopsy is usually unnecessary

Treatment

- Supportive therapy
- Minimize metabolic disturbance and other complications
- Preparation of renal replacement therapy if approaching stage 5 CKD

Tubulointerstitial Diseases

- Anatomically, comprises >95% of the kidney
- Acute TIN may cause important renal dysfunction;
~5-15% acute renal failure cases
- Left unchecked, chronic TIN is the major pathway leading to chronic kidney disease



Clinical manifestations

Due to interstitial inflammation & tubular dysfunction

- Loss of renal function
- Abnormal urine sediment, e.g. WBCs, and proteinuria
- Electrolyte disturbance
- Acid-base disorders

Entities

- Acute interstitial nephritis
 - Drugs, infections, autoimmune disease
- Chronic interstitial nephritis
 - Etiology
- K⁺-wasting tubular disorders
- Renal tubular acidosis

Acute kidney injury

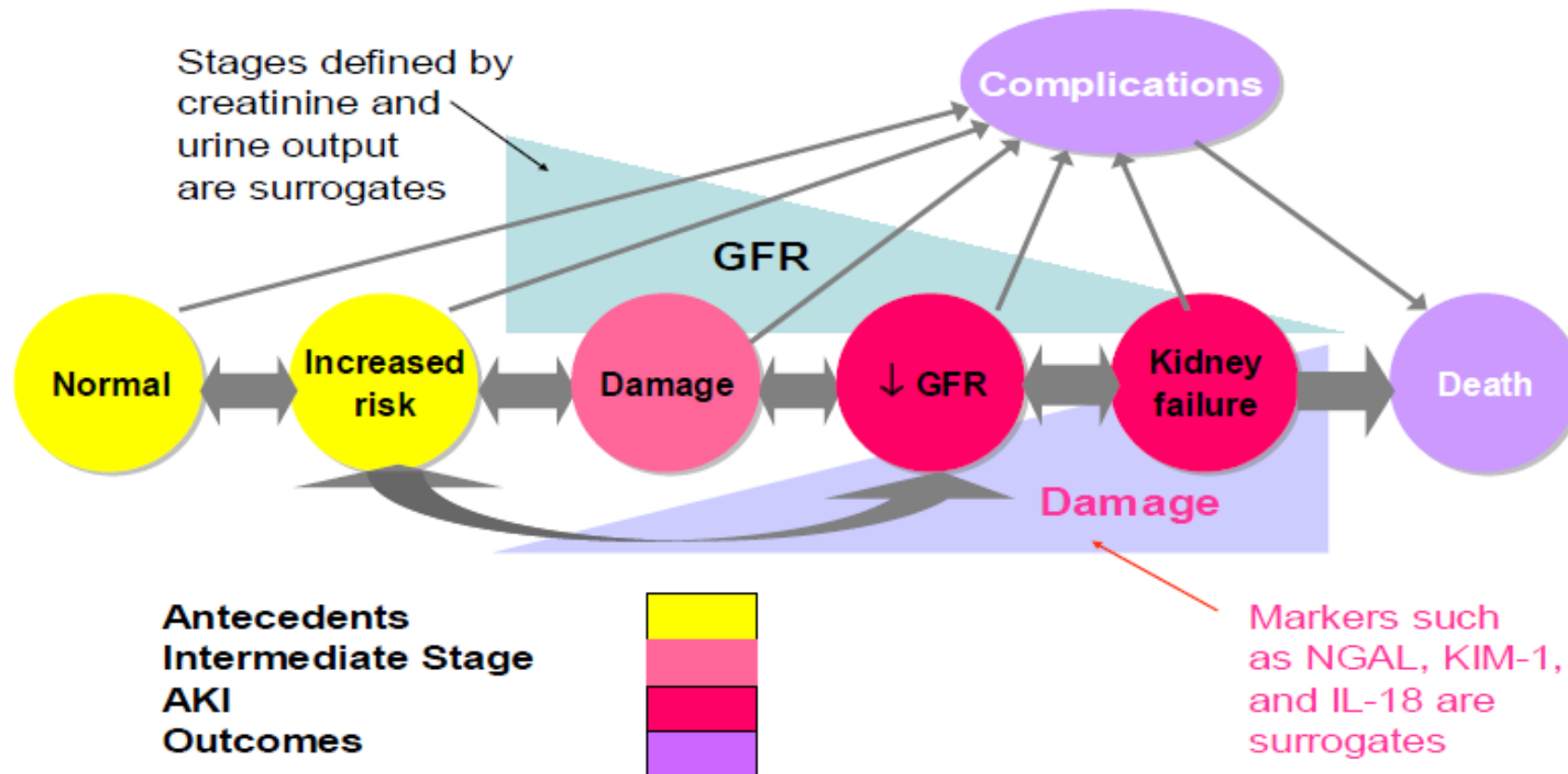
- AKI is defined by an abrupt decrease in kidney function
- Various etiologies – ATN, acute interstitial nephritis, acute glomerular and vasculitic renal diseases, prerenal azotemia, and acute postrenal obstructive nephropathy
- More than one of these conditions
- Even mild, reversible AKI has important clinical consequences, e.g. CKD transition
- Akin to acute lung injury or acute coronary syndrome

AKI-old definitions (ungraded)

- Clinical:
 - abrupt decline in glomerular filtration rate (GFR) over hours or days
 - No unified consensus on absolute definition
- Biochemical:
 - Rise in SeCr by > 50 $\mu\text{mol/l}$ over baseline
 - 50% increase in SeCr
 - Reduction in CrCl by $> 50\%$
 - Rise from normal to 180-270 $\mu\text{mol/L}$
 - Doubling of SeCr
 - Rise of SeCr by 26.5 $\mu\text{mol/L}$ over 24 h
 - u/o < 0.5 ml/kg/h for > 6 h

Stage	Serum creatinine
1	1.5-1.9 times baseline OR ≥ 0.3 mg/dL (≥ 26.5 μ mol/L) increase
2	2.0-2.9 times baseline
3	3.0 times baseline OR Increase in serum creatinine to ≥ 4.0 mg/dL (≥ 353.6 μ mol/L) OR Initiation of renal replacement therapy, OR in patients < 18 years, decrease in eGFR to < 35 mL/min/1.73 m ²

Conceptual model for AKI



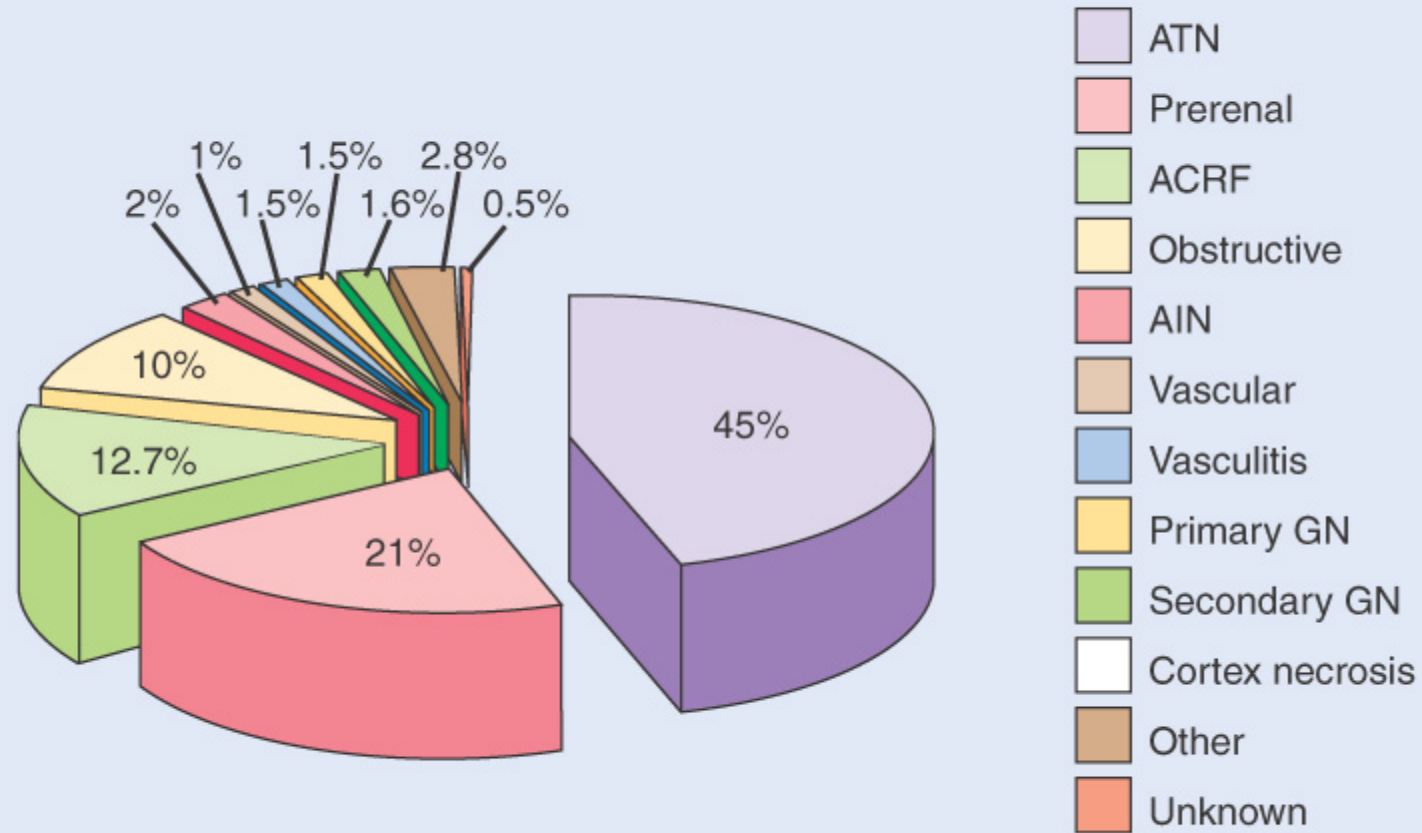
Risk factors

- Preexisting CKD
- Diabetic kidney disease
- Heart failure
- Liver disease
- Hypovolemia
- Age > 50 yrs
- Sepsis
- Trauma
- Postoperative (e.g. post-CABG)
- Cancer

Aetiology

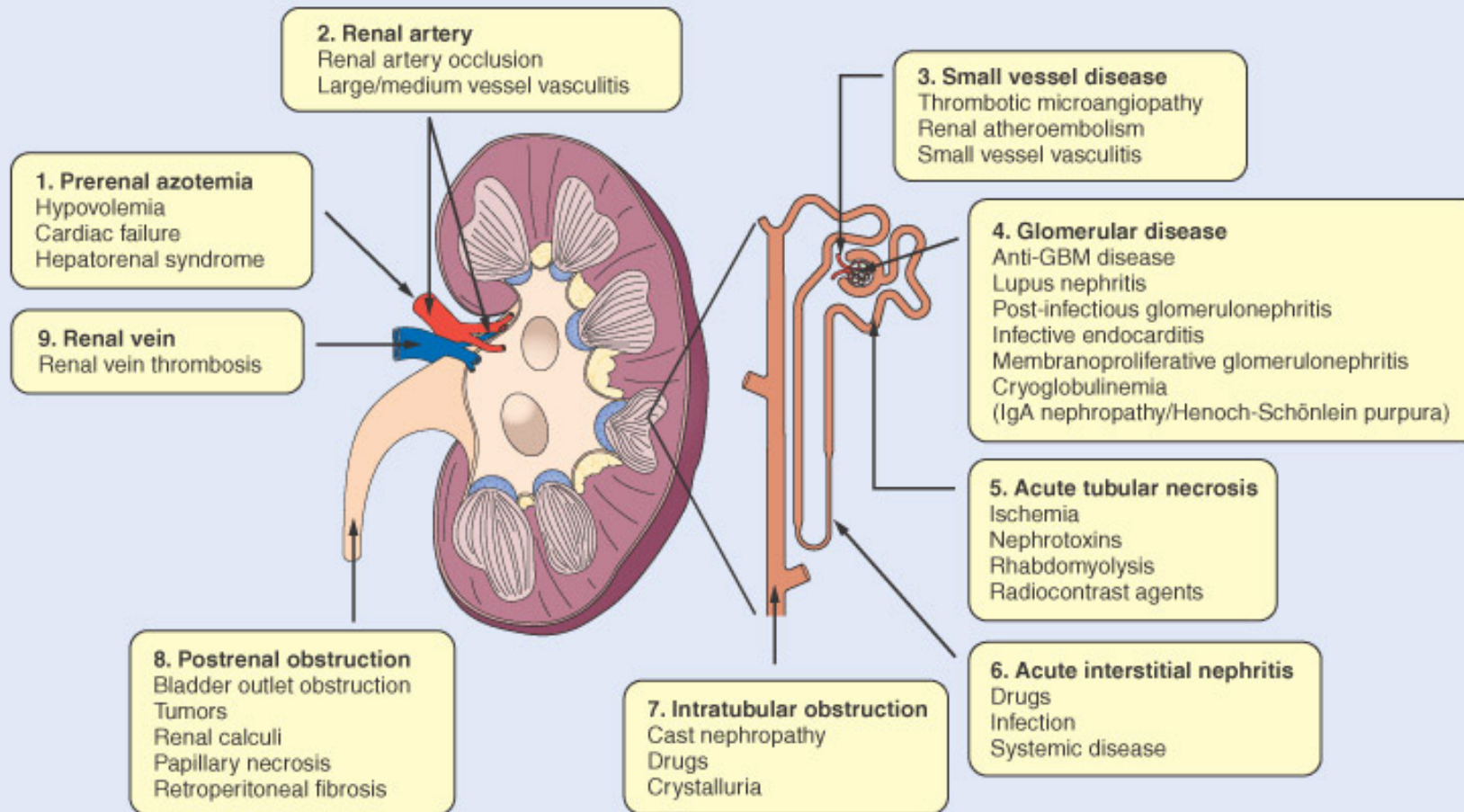
- Pre-renal ($>1/2$)
- Intrinsic ($<1/2$)
 - Tubular necrosis
 - Ischaemia
 - Toxins
 - Interstitial nephritis
 - Acute Gn (RPGn)
- Post-renal ($<10\%$)

Most important causes of community-acquired ARF



(Redrawn from Liano F, Pascual J: Epidemiology of acute renal failure: A prospective, multicenter, communitybased study. Madrid Acute Renal Failure Study Group. Kidney Int 1996; 50:811-818.)

Causes of AKI



21中成藥含馬兜鈴酸回收

【明報專訊】衛生署昨日起禁止含可致腎衰竭及癌症的馬兜鈴酸中成藥銷售及入口，並點名回收21種中成藥，包括知名藥廠「馬百良」、「位元堂」及「北京同仁堂」的產品。不過有業界人士指被點名回收的全是自願送檢的中成藥，坊間有數以百計未送檢但含馬兜鈴酸的中成藥，仍在售賣。本報昨日在市面買到標明含細辛（有馬兜鈴酸）的中成藥。

包括馬百良位元堂北京同仁堂產品

衛生署發言人亦承認坊間可能仍有含馬兜鈴酸的中成藥存在。該署是在較早前由藥廠及代理商送交檢驗約100種中成藥中，驗出21種藥含馬兜鈴酸並宣布回收。發言人強調，當局會加強在市面抽驗其他藥品，一經發現有馬兜鈴酸，會即時公布及回收。

可致腎功能衰竭尿道癌

馬兜鈴酸可導致急性或慢性腎功能衰竭及尿道癌。衛生署早在4月宣布，6月1月起（昨日）禁止銷售所有含馬兜鈴酸的中成藥，將來亦不會批准此類中成藥註冊。

然而關注香港中醫藥政策聯席會秘書長潘光濤透露，昨剛收到衛生署邀請，明日才開會，討論如何規管含馬兜鈴酸中藥，他批評該署先禁售部分中成藥，才討論如何規管，做法是「打死狗講價」，對業界不公平。

衛生署昨點名21種中成藥含馬兜鈴酸，指令製造商即時回收（詳見右圖）。該署呼籲市民即時停服有關藥物，銷毀或交回中成藥的代理商，或於辦公時間交回九龍南昌街382號公共衛生檢測中心2樓衛生署中藥事務組。

業界料300中成藥「漏網」

不過有被點名的中成藥代理商對衛生署的做法不滿，並指今次是「自投羅網」，因為今次被點名的中成藥是較早前響應衛生署呼籲交出產品作免費檢驗，反觀坊間現時仍有300多種含馬兜鈴酸中成藥（由商會評估），仍可繼續

含馬兜鈴酸 要回收的中成藥

中成藥名稱(製造商)	香港代理商 聯絡電話	中成藥名稱(製造商)	香港代理商 聯絡電話
海瓶牌複方三蛇膽川貝末 (成都九芝堂金鼎藥業(原成都中藥廠))	2743 3416	鼻敏寧(同德堂藥廠)	2545 1551
順氣化痰止咳丸(廣州中一藥業)	2393 2989	小青龙湯顆粒	2753 8818
細辛中藥配方顆粒 (培力(南寧)藥業)	2840 1840	苓甘薑味辛夏仁湯顆粒 (順天堂藥廠股份公司)	3428 2333
益腎蠲痺丸(江蘇清江藥業)	2511 3707	麻黃附子細辛湯顆粒 (順天堂藥廠股份公司)	3428 2333
濟生堂鼻敏清(濟生堂製藥廠)	2895 2018	補肺阿膠散顆粒 (順天堂藥廠股份公司)	3428 2333
護齒靈(四川省中藥廠)	2771 1279	龍膽瀉肝丸(河南時珍製藥)	2571 0775
三癆湯(培力(南寧)藥業)	2840 1840	龍膽瀉肝湯顆粒	2753 8818
鼻敏通(位元堂藥廠)	2727 8911	杜仲筋骨痺痛丸(中國貴陽中藥廠)	2562 6129
風濕丸(李熊記)	2541 2080	追風透骨丸 (信興藥粉、寶和堂製藥廠)	2505 7336
白鳳丸(李熊記)	2541 2080	蘇合丸(馬百良廠)	2544 0539
順氣止咳化痰丸 (北京同仁堂科技發展股份製藥廠)	2587 8788		

衛生署查詢電話：2319 5119（辦公時間）

資料來源：衛生署 註：馬兜鈴酸可導致急性或慢性腎功能衰竭及尿道癌

「漏網」出售。

該代理商解釋，被點名的中成藥即使稍後更改成分，銷路一定會受打擊，但現時「漏網」的中成藥，可以有時間更改製藥成分，改用不含馬兜鈴酸成分的中藥，甚至不列出有問題的配方，衛生署及市民根本難以察覺。

「現時坊間有逾千種中成藥，政府根本難以全面檢驗。

」被點名的廠家稱。

醫學會會長勞永樂認為，廠家應向衛生署提供資料，由衛生署跟進。

本報昨日在藥房仍購得在包裝及說明書列明含細辛的「馬百良追風蘇合丸」。據中醫師潘光濤說，細辛含有馬兜鈴酸。

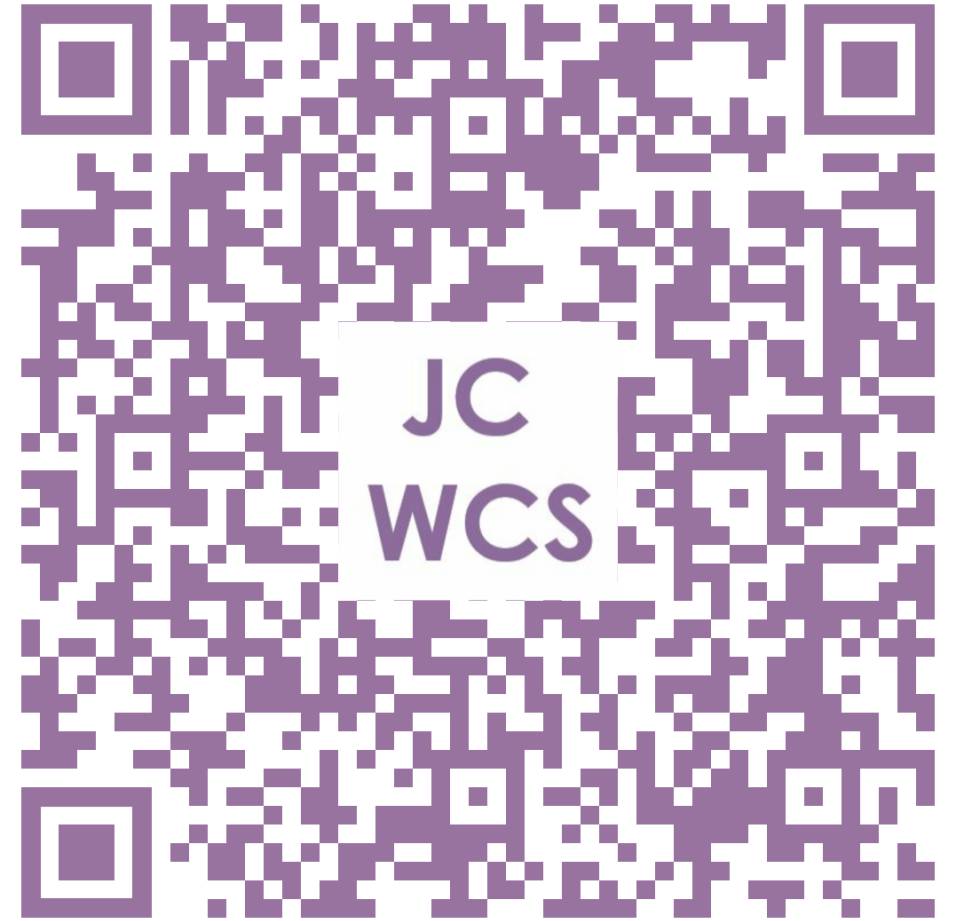
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JC Whole Class Session

YOU THINK

THANK

YOU!



Look for today's teacher
and the others you may have missed.
(HKU Portal login required)